

THE UNIVERSITY OF HONG KONG FACULTY OF BUSINESS AND ECONOMICS

MSBA7017 Financial Engineering

GENERAL INFORMATION

Instructor: Prof. Liao Wang

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Consultation times: By appointment

Tutor: TBA

Pre-requisites: MSBA7002 Business Statistics and MSBA7003 Decision Analytics

Co-requisites: N/A

Mutually exclusive: N/A

Course Website: See Moodle.

Textbook:

Notes and slides will be distributed and used as the main class materials. Supplementary readings will also be assigned.

The following books are recommended references but not required.

[L] Luenberger, D.G., Investment Science, Oxford University Press, New York, 1998.

[H] Hull, J.C., Options, Futures, and Other Derivatives, latest ed., Prentice Hall, Upper Saddle River, NJ.

[J] An Introduction to Statistical Learning with Applications in R (Springer) by James, Witten, Hastie and Tibshirani.

COURSE DESCRIPTION

The goal of this course is twofold. The primary objective is to equip students with analytical foundation and hands-on skills in solving investment and trading problems. Implementation of the analytics (e.g., building term structure models, portfolio construction, pricing options by simulation, etc.) by developing models in spreadsheet/VBA or Python will also be emphasized. The secondary objective is to introduce Financial Technology ("FinTech") --- one of the most promising sectors in financial industry. We will have an overview on the active product areas in FinTech, including capital raising platforms and digital wealth management. Digital currency and the closely related technology (blockchain) will also be highlighted.

COURSE OBJECTIVES

This course introduces basic methods and tools in financial engineering, and provides an overview on Financial Technology industry. Upon completing this course, students should:

1. Be fluent in applying basic models/methods that are widely used in financial industry;
2. Be able to quantitatively analyze investment and trading problems, and
3. Build spreadsheet/VBA or Python models from financial data to provide concrete solutions;
4. Possess overall understanding and develop business acumen for FinTech.

Programme Learning Outcomes

PLO1: Acquisition and internalization of knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES			
Course Learning Outcomes		Aligned Programme Learning Outcomes	
CLO1 Select appropriate and effective model to clearly define and formulate problems encountered in real investment and trading activities.		CLO1-PLO1, PLO2	
CLO2 Adept in applying the major methods learned to solve the problems.		CLO2-PLO1, PLO2	
CLO3 Demonstrate proficiency in implementing the models/methods to financial data using spreadsheet/VBA modeling or Python.		CLO3-PLO2	
CLO4 Understand the economics of FinTech; possess knowledge of the products and solutions, as well as technology used in this industry.		CLO4-PLO1, PLO4	
CLO5 Effectively communicate ideas, clearly explain solution approaches, and interpret the results with insights.		CLO5-PLO3, PLO5	
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)
T&L1. Lectures (present concepts, theories, and examples)		30	25%
T&L2. Tutorials (problem sets, spreadsheet/VBA/Python coaching, and interactive discussions)		12	10%
T&L3. Bi-weekly assignments		36	30%
T&L4. Self-study		42	35%
Total		120	100%
Assessment Methods			
Brief Description (Optional)		Weight	Aligned Course Learning Outcomes
A1. Assignments	Each assignment will comprise of: 1. One coding problem 2. 2-3 computational problems. 3. 2-3 questions based on assigned reading.	40%	CLO1-4
A2. Final Exam		50%	CLO1-4
A3. In-class participation		10%	CLO5
Total		100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	Candidate demonstrates strong understanding of all relevant knowledge and can apply appropriate method to each specific problem. Capable in spreadsheet implementation. Active in class and contribute meaningful discussions.		
B+, B, B-	Candidate demonstrates good understanding of all relevant knowledge and tends to apply the methods learnt appropriately. Can use spreadsheet to build models, but might need guidance. Active in class.		

C+, C, C-	Candidate has adequate understanding of course content, but cannot apply the knowledge appropriately. Shows effort in spreadsheet modelling, but has major difficulty.
D+, D	Candidate shows some effort, but only has a minimum understanding of the course content.
F	Candidate does not show any effort.
Assessment Rubrics for Each Assessment	
Assessment Rubrics for Assignments	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge. • Strong implementation skills reflected by clean and accurate spreadsheet modeling. • Accurate in analytical/computational problems. • Answers to questions based on readings show strong understanding of the materials and insightful thinking.
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge. • Good implementation skills reflected by a completed spreadsheet modeling that addresses the problem. • Able to work out analytical/computational problems to an adequate extent. • Answers to questions based on readings reflect good understanding of the materials.
C+, C, C-	• Demonstrate an adequate understanding of the course materials. • Basic implementation skills reflected by correct approach to spreadsheet modelling but not able to complete. • Have difficulty in working out analytical/computational problems but show some conceptual understanding. • Answers to questions based on readings show good efforts and some understanding, but cannot grasp the core ideas.
D+, D	• Demonstrate a minimum understanding of the course materials. • Obvious lack in implementation skills. • Major difficulty in working out analytical/computational problems. • Answers to questions based on readings show some efforts but the understanding is wrong.
F	• Demonstrate a poor understanding of the concepts involved. • No effort is shown.
Assessment Rubrics for Final Exam	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge. • Accurate in analytical/computational problems. • Answers are direct to the point with conciseness.
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge. • Able to work out analytical/computational problems, but with minor/moderate mistakes.

C+, C, C-	- Demonstrate an adequate understanding of course materials. - Have some difficulty in working out analytical/computational problems but show conceptual understanding.
D+, D	- Demonstrate a minimum understanding of the concepts involved. - Major difficulty in working out analytical/computational problems with occasional conceptual mistakes.
F	- Demonstrate a poor understanding of the concepts involved. - Answers do not address the problems/questions.

Assessment Rubrics for In-Class Participation

A+, A, A-	- Active in both starting and participating in discussion. - Demonstrate a strong understanding of all relevant knowledge. - Handling questions professionally. - Contribute meaningful discussions.
B+, B, B-	- Active in both starting and participating in discussion. - Demonstrate a good understanding of all relevant knowledge. - Handling questions professionally.
C+, C, C-	- Some participation in the discussion. - Demonstrate an adequate understanding of course materials.
D+, D	- Minimum or no participation in the discussion. - Demonstrate a minimum understanding of the concepts involved.
F	- Demonstrate a poor understanding of the concepts involved. - No participation.

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Week	Lecture	Lecture	Topic
1			- Introduction to financial engineering: problem categories, methods and objectives. - Bonds: price-yield relation; yield curve; spot rate and spot rate term structure; forward rate and forward rate curve; durations and immunization. - FinTech: introduction, history and fintech hubs.
2			- Bonds (ct'd): numerical examples. - Portfolio optimization: risk and return basics; Markowitz mean-variance models; risk-return efficient frontier; - FinTech: crowdfunding and market place lending.
3			- Portfolio optimization (ct'd): Capital Asset Pricing Model (CAPM); security market line (SML), beta, alpha; numerical examples. - Portfolio optimization (ct'd): Black-Litterman Model; numerical examples. - FinTech: digital wealth management

4			- Option theory: option basics; payoff functions; no-arbitrage arguments; common option strategies; Black-Scholes formula; risk-neutral pricing. - FinTech: Bitcoin and blockchain.
5			- Option theory (ct'd): delta-hedging; greeks; numerical examples. - FinTech: Bitcoin and blockchain (ct'd). - Machine learning: (review) linear classifier and support vector machine (svm); application of ML in financial services. - Handling big data: MapReduce and distributed file system. - Final Exam Review; Q&A.
Exam	See Teaching Plan		

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

See "textbook" section at the beginning. Others to be determined.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE


Online response via Moodle site

SETL around the end of the semester
(please specify)



Others: _____

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Scholastic dishonesty also includes, but is not limited to, providing false or misleading information to receive a postponement or an extension on an exam or other assignment. The responsibilities of both students and faculty with regard to scholastic dishonesty are described in detail in the Disciplinary Committee Regulations (<http://www4.hku.hk/pubunit/calendar/2016-2017/a/c/32-2016-2017/400-disciplinary-committeeregulations>). By teaching this course, I have agreed to observe all of the faculty responsibilities described in that document. By enrolling in this class, you have agreed to observe all of the student responsibilities described in that document. If the application of that policy statement to this class and its assignments is unclear in any way, it is your responsibility to ask me for clarification.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Late submission of assignments will invoke a 20% penalty (of that particular assignment grade) for each day after the due date (thus the grade will be 0 after 5 days).